§4.3 The Fundamental Theorem for Line Integrals

Review of the Fundamental Theorem of Calculus:

Theorem. Fundamental Theorem of Calculus (Review)

If $F'(x)$ is continuous on the interval $[a, b]$, then

$$\int_a^b F'(x)dx = F(b) - F(a)$$

Let C be a curve defined vector function $\vec{r}(t)$, $a \le t \le b$.

Let $\vec{F}$ be a vector field (on $\mathbb{R}^2$ or $\mathbb{R}^3$) defined on C.

Then the line integral of $\vec{F}$ along C is defined as $\int_C \vec{F} \cdot d\vec{r}$.

The Fundamental Theorem for line integrals.

Theorem. Fundamental Theorem for line integrals

If C is smooth and $\vec{F}$ is conservative ($\vec{F} = \nabla f$), then

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a)).$$

This means that, in this nice situation, we can evaluate the line integral using only value at the endpoints of the potential function f.

proof:

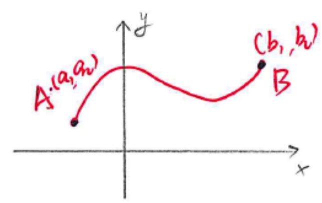
$$\int_C \nabla f \cdot d\vec{r} = \int_a^b \nabla f \cdot r'(t)\, dt$$

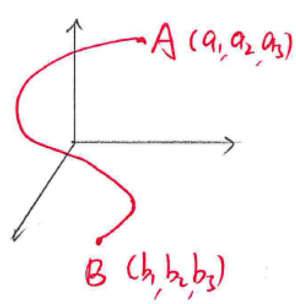
$$= \int_a^b \langle f_x, f_y \rangle \cdot \langle x'(t), y'(t) \rangle\, dt$$

$$= \int_a^b f_x \cdot x'(t) + f_y \cdot y'(t)\, dt$$

$$= \int_a^b \frac{d}{dt}\left(f(\vec{r}(t))\right) dt$$

$$= f(\vec{r}(b)) - f(\vec{r}(a))$$

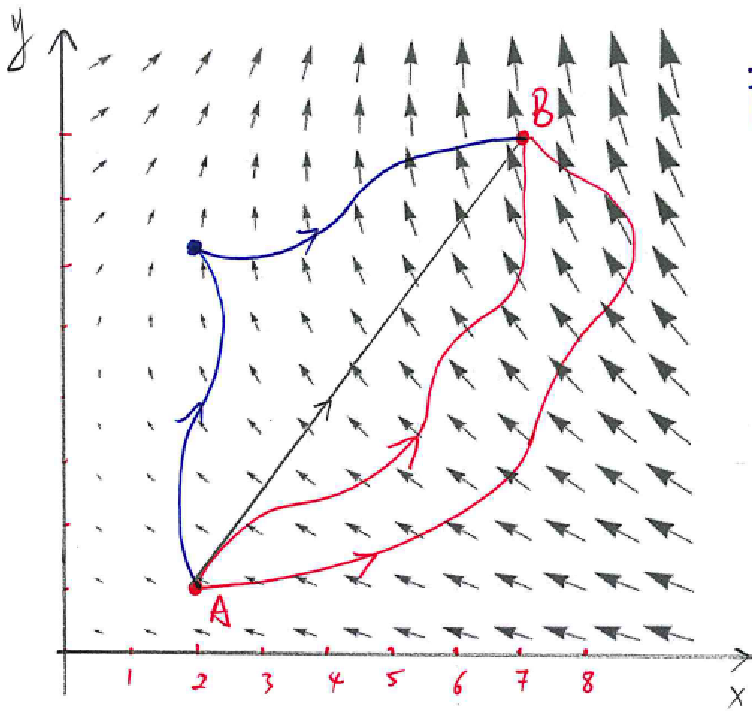




If $\vec{F} = \nabla f$ is defined on a region D and C_1, C_2 are two (piecewise) smooth curves on D with the same initial and terminal points, then

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$$

Line integrals of conservative vector fields are independent of path.



$\vec{F}(x, y) = \langle y^2 - 3x^2, 2xy \rangle$

$f(x,y) = y^2 x - x^3$

$\vec{F} = \nabla f$

$A = (2, 1)$

$B = (7, 8)$

$$\int_C \vec{F}\, d\vec{r} = f(7,8) - f(2,1) = 105 - 6 = 99$$

for any C from A to B.
good curve

Example 1. (Gravitational Field) Let $\vec{x} = \langle x, y, z\rangle \in \mathbb{R}^3$. The gravitational force acting on the object at $\vec{x}$ is

$$\vec{F}(\vec{x}) = -\frac{mMG}{|\vec{x}|^3}\vec{x}$$

m and M are masses of the two objects. $G = 6.67408 \times 10^{-11}$ is the universal Gravitational constant.

The Gravitational Field is a conservative vector field, $\vec{F} = \nabla f$, for

$$f(x, y, z) = \frac{mMG}{\sqrt{x^2 + y^2 + z^2}}$$

Problem: Find the work done by $\vec{F}$ in moving a particle with mass m from point $(1, 0, 0)$ to $(1, 2, 9)$.

By fundamental theorem for line integral

$$\text{work} = \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_C \nabla f \cdot d\vec{r}$$

$$= f(1,2,9) - f(1,0,0)$$

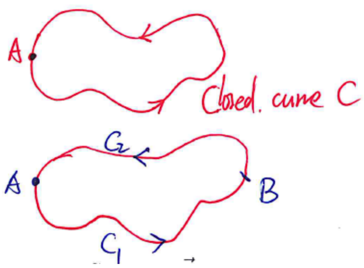
$$= \frac{mMG}{\sqrt{1+4+81}} - \frac{mMG}{\sqrt{1}}$$

$$= \left(\frac{1}{\sqrt{86}} - 1\right) mMG$$

A curve is **closed** if its terminal point coincides with its initial point.

Theorem.

The line integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path in D if and only if $\int_C \vec{F} \cdot d\vec{r} = 0$ for every closed path C in D.



$$\int_C \vec{F} d\vec{r} = \int_{C_1} \vec{F} d\vec{r} + \int_{C_2} \vec{F} d\vec{r}$$

$$= \int_{C_1} \vec{F} d\vec{r} - \int_{-C_2} \vec{F} d\vec{r}$$

$$= 0$$

Suppose $\vec{F}$ is **continuous** on an **open**, **connected** region D.

Theorem.

The line integral $\displaystyle\int_C \vec{F} \cdot d\vec{r}$ is independent of path in D if and only if $\vec{F}$ is conservative on D, that is, $\vec{F} = \nabla f$.

Question:

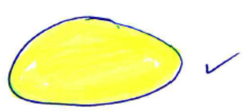
How to determine whether or not a vector field $\vec{F}$ is conservative?

Theorem.

If $\vec{F}(x, y) = P(x, y)\vec{i} + Q(x, y)\vec{j}$ is conservative on D, then

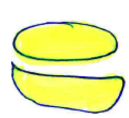
$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

A region D in the plane $\mathbb{R}$ is **simply-connected** if it is connected and every simple closed curve in D enclose only points in D.



NOT simply-connected:

connected



not connected

Theorem.

Let D be a **simply-connected**, **open** region.

Let $\vec{F}(x,y) = P(x,y)\vec{i} + Q(x,y)\vec{j}$ be a vector field such that P and Q have continuous first derivatives and

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

on D, then $\vec{F}$ is conservative.

Example 2. Determine whether or not each of the vector field is conservative.

(1) $\vec{F}(x,y) = (2x+y)\vec{i} + (x+2y)\vec{j}$

$\frac{\partial P}{\partial y} = 1$

$\frac{\partial Q}{\partial x} = 1$

$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow$ Conservative

(2) $\vec{F}(x,y) = \langle 2x - y, x+1 \rangle$

$\frac{\partial P}{\partial y} = -1$

$\frac{\partial Q}{\partial x} = 1$

$\Rightarrow \frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x} \Rightarrow$ Not conservative

(3) $\vec{F}(x,y) = \langle 4 + 2xy, x^2 + y^2 \rangle$

$\frac{\partial P}{\partial y} = 2x$

$\frac{\partial Q}{\partial x} = 2x$

$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow$ Conservative.

Example 3. Let $\vec{F}(x, y) = \langle 4 + 2xy, x^2 + y^2 \rangle$.

(1) Find a function f such that $\nabla f = \vec{F}$.

$$\nabla f = \langle f_x, f_y \rangle = \vec{F} = \langle 4+2xy, x^2+y^2 \rangle$$

So $f_x = 4 + 2xy \Longrightarrow f(x,y) = 4x + x^2 y + g(y)$

$f_y = x^2 + y^2$ $\qquad f_y = x^2 + g'(y)$

$\Rightarrow x^2 + y^2 = x^2 + g'(y) \Rightarrow g'(y) = y^2 \Rightarrow g(y) = \frac{y^3}{3} + K$

So $f(x,y) = 4x + x^2 y + \frac{y^3}{3} + K$

(2) Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$, where C is the curve defined by $\vec{r}(t) = \langle e^t \cos t, e^t \sin t \rangle$ for $0 \le t \le \pi/2$.

- $\vec{r}(0) = \langle 1, 0 \rangle$ initial
- $\vec{r}(\frac{\pi}{2}) = \langle 0, e^{\frac{\pi}{2}} \rangle$ terminal

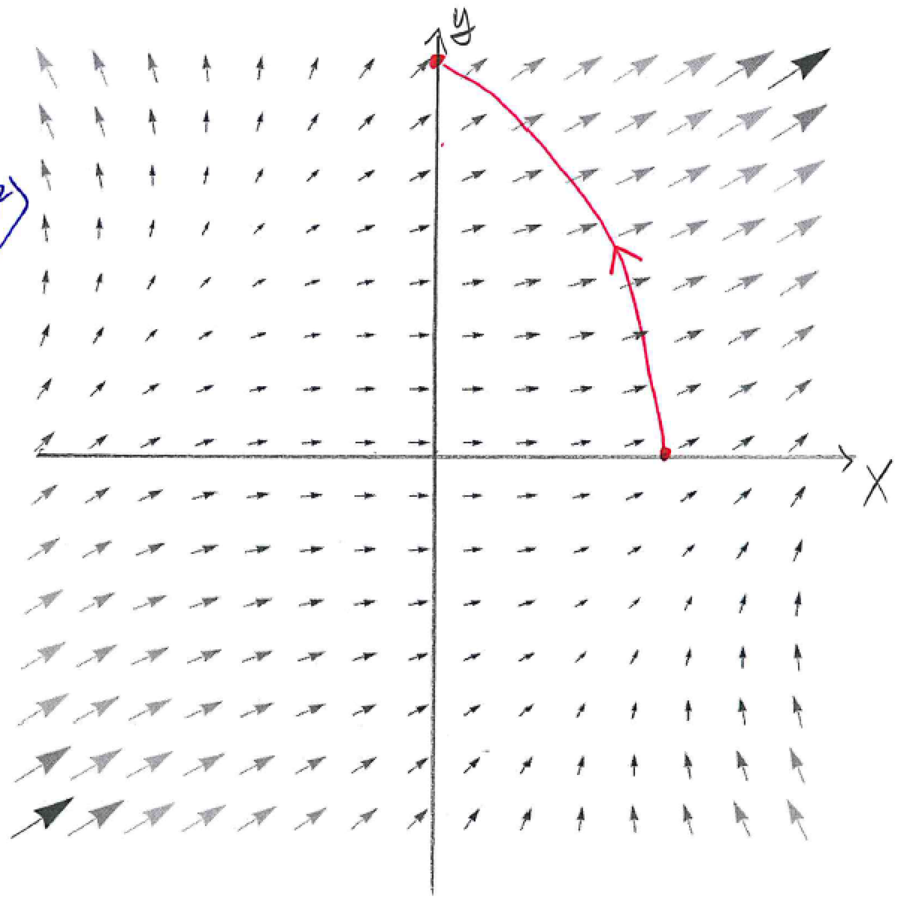
$$\int_C \vec{F}\, d\vec{r} = f(0, e^{\frac{\pi}{2}}) - f(1,0)$$

$$= \frac{(e^{\frac{\pi}{2}})^3}{3} - 4$$

$$= \frac{e^{\frac{3\pi}{2}}}{3} - 4$$

Example 3.

$\vec{F} = \langle 4+2xy, x^2+y^2 \rangle$



Example 4. Let $\vec{F}(x,y) = \langle e^y + \cos y, xe^y - x\sin y \rangle$.

(1) Determine whether or not $\vec{F}$ is conservative.

$$\frac{\partial P}{\partial y} = e^y - \sin y$$
$$\frac{\partial Q}{\partial x} = e^y - \sin y$$
$$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \vec{F} \text{ is conservative.}$$

(2) Find a function f such that $\nabla f = \vec{F}$.

$$\begin{cases} f_x = e^y + \cos y \\ f_y = xe^y - x\sin y \end{cases}$$
$$\Rightarrow f = xe^y + x\cos y + g(y)$$
$$\Downarrow$$
$$f_y = xe^y - x\sin y + g'(y)$$
$$g'(y) = 0 \Rightarrow g(y) = k$$
$$\Rightarrow f = xe^y + x\cos y + k$$

(3) Use part (2) to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is a curve from $(1,0)$ to $(0,3)$

$$\int_C \vec{F} \cdot d\vec{r} = f(0,3) - f(1,0) = 0 - (1+1) = -2$$

Example 5. Let $\vec{F}(x,y) = (6x^2y + 2x\ln y)\vec{i} + (2x^3 + \dfrac{x^2}{y})\vec{j}$.

(1) Determine whether or not $\vec{F}$ is conservative.

$$\frac{\partial P}{\partial y} = 6x^2 + \frac{2x}{y}$$

$$\frac{\partial Q}{\partial x} = 6x^2 + \frac{2x}{y}$$

$$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \text{conservative}$$

(2) Find a function f such that $\nabla f = \vec{F}$.

$$\begin{cases} f_x = 6x^2y + 2x\ln y \\ f_y = 2x^3 + \dfrac{x^2}{y} \end{cases}$$

$$f_x = 6x^2y + 2x\ln y \Rightarrow f(x,y) = 2x^3y + x^2\ln y + g(y)$$

$$\Downarrow$$

$$f_y = 2x^3 + \frac{x^2}{y} + g'(y)$$

$$g'(y) = 0 \Rightarrow g(y) = k$$

So $f(x,y) = 2x^3y + x^2\ln y + k$

(3) Use part (2) to evaluate $\int_C \vec{F}\cdot d\vec{r}$ where C is a curve from $(1,1)$ to $(0,4)$.

$$\int_C \vec{F}\cdot d\vec{r} = f(0,4) - f(1,1) = 0 - (2+0) = -2$$

Example 6. Let $\vec{F}(x,y,z) = \langle yz, xz, xy + e^z \rangle$.

(1) Find a function f such that $\nabla f = \vec{F}$.

$$\begin{cases} f_x = yz \\ f_y = xz \\ f_z = xy + e^z \end{cases} \qquad \langle f_x, f_y, f_z \rangle$$

So $f = xyz + e^z + K$

$$f = xyz + \underline{g(y,z)}$$

$$\Rightarrow \begin{cases} f_y = xz + g_y \\ f_z = xy + g_z \end{cases} \Rightarrow \begin{matrix} g_y = 0 \\ g_z = e^z \end{matrix} \Rightarrow g(y,z) = e^z + K$$

(2) Use part (1) to evaluate $\int_C \nabla f \cdot d\vec{r}$ where C is a line from $(1,2,0)$ to $(0,3,1)$

$$\int_C \nabla f \cdot d\vec{r} = f(0,3,1) - f(1,2,0)$$

$$= e - (e^0)$$

$$= e - 1$$